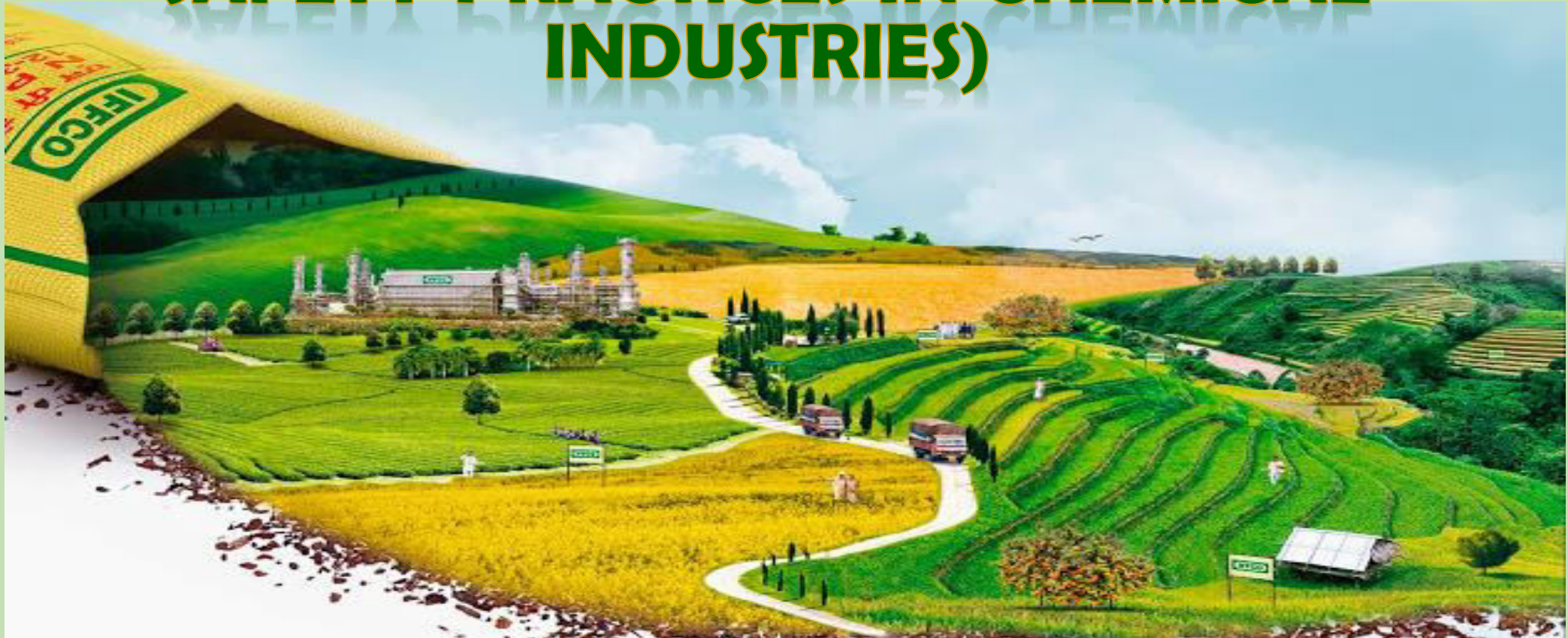


INDIAN FARMERS FERTILIZER COOPERATIVE LIMITED

PARADEEP UNIT

**CHEMICAL SAFETY (ENGINEERING
INTERVENTIONS ON SAFETY & BEST
SAFETY PRACTICES IN CHEMICAL
INDUSTRIES)**



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Dy. General Manager
Technical Services
IFFCO PARADEEP



Good safety culture results in Injury free smiling people, healthy work environment and good quality products

Methodology to Ensure Safety at Workplace

- ❖ PRO ACTIVE ACTION
- ❖ PREVENTION
- ❖ MITIGATION

PRO ACTIVE ACTION : By reducing the frequency of
man and machine interaction

Automatic Lubrication System in Conveyor Belts

- ❖ This system perform automatic lubrication in running condition.
- ❖ Earlier manual lubrication involved manpower which may result in accidents.
- ❖ To avoid this, we have installed automatic lubrication which do not require manpower.



Lubrication Pump



Central Distribution Unit



Automatic Lubrication System in Conveyor Belts

Benefits of Automatic Lubrication System-

- Enhance safety
- Proper and controlled lubrication
- No contamination of lubricant
- Saving in cost of lubrication
- Significant reduction in downtime
- Increased bearing life

Self Adjusting Automatic Belt Trackers in Sulphur Conveyor

- ❖ Traditionally manual belt trackers were risky to adjust during running condition.
- ❖ To ensure safety during operation, automatic belt tracker has been installed. It also ensures zero spillage caused due to belt sway.



Manual belt trackers



Self adjusting belt trackers

Self Adjusting Automatic Belt Trackers in Sulphur Conveyor

Benefits of Self adjusting automatic belt trackers-

- Enhanced safety during running operation
- Alignment of belt is ensured
- No spillage of product
- No damage to belt and nearby structure
- Significant reduction in downtime of conveyor

Use of HPPE Rollers

- ❖ HPPE stands for **High performance polyethylene**.
- ❖ HPPE is an alternate to MS & Rubber lagged Idlers.
- ❖ Earlier used MS & Rubber lagged idlers were prone to produce sparks due to friction and lead to fire incidents.



HPPE Roller

Before Modification



Steel Carrying Idler



MS with R/L Impact Idler

After Modification



HPPE Carrying Idler



HPPE Impact Idler

Use of HPPE Rollers

Benefits of HPPE rollers-

- Enhanced safety during running operation
- No risk of fire in belts
- Lighter and safe for handling
- Lower noise
- Lesser power on start-up
- Non corrosive shell ideal for corrosive environment

Dedusting System for Conveyors

- ❖ Conventional type de-dusting systems were provided to take care of dust emission from conveyors at transfer points.
- ❖ They were not effective due to high capacity of conveyor (3200 MTPH)



Top skid mounted dedusting system

Extraction point

Old conventional type de-dusting system

Dedusting System for Conveyors

- Modified skirt board & sealed hood type de-dusting systems are provided so that dust generated during transfer of material from one conveyor to another can be effectively sucked by an external ID Fan equipped with bag filters by creating negative draft.

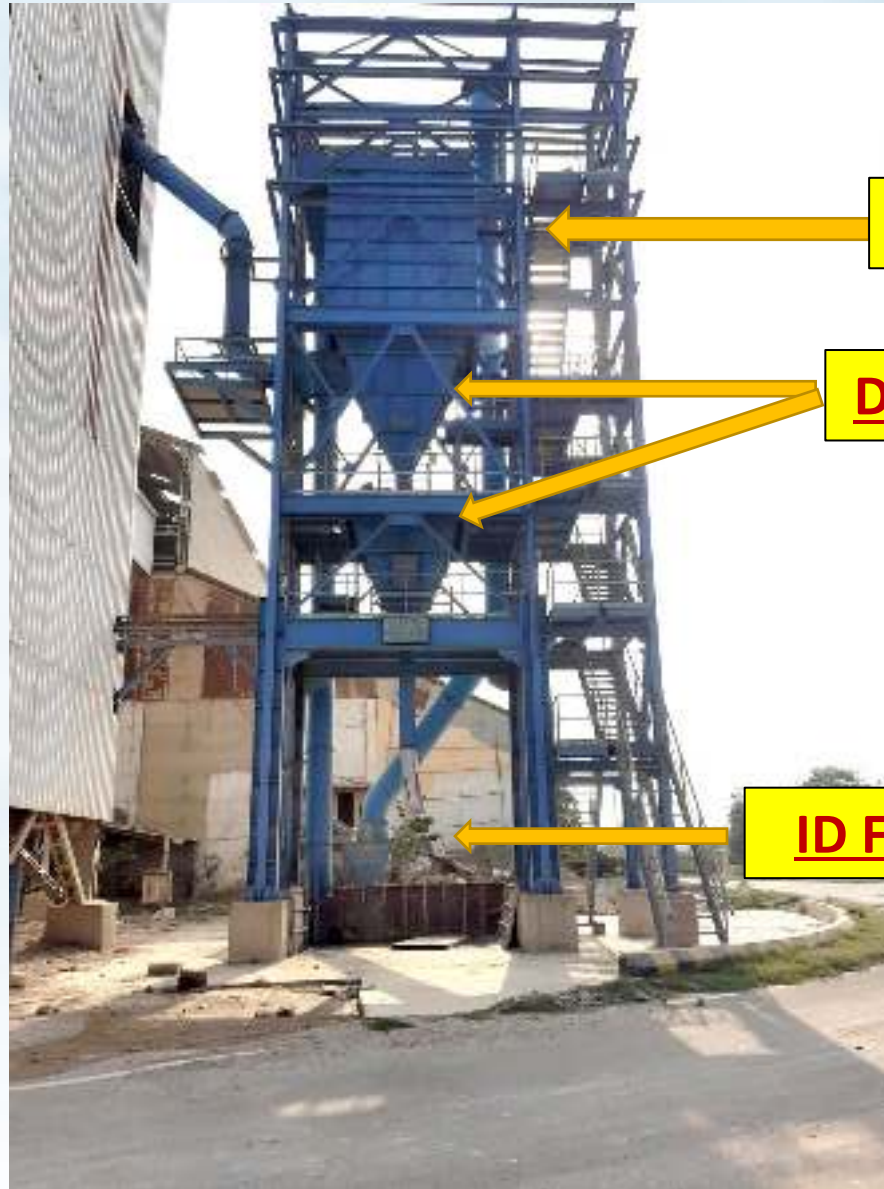


Modified skirt board & sealed hood type



Dust extraction point from conveyor

Dedusting System for Conveyors



Dust separated in bag filter

Dust collected in hopper

ID Fan

Dedusting System for Conveyors

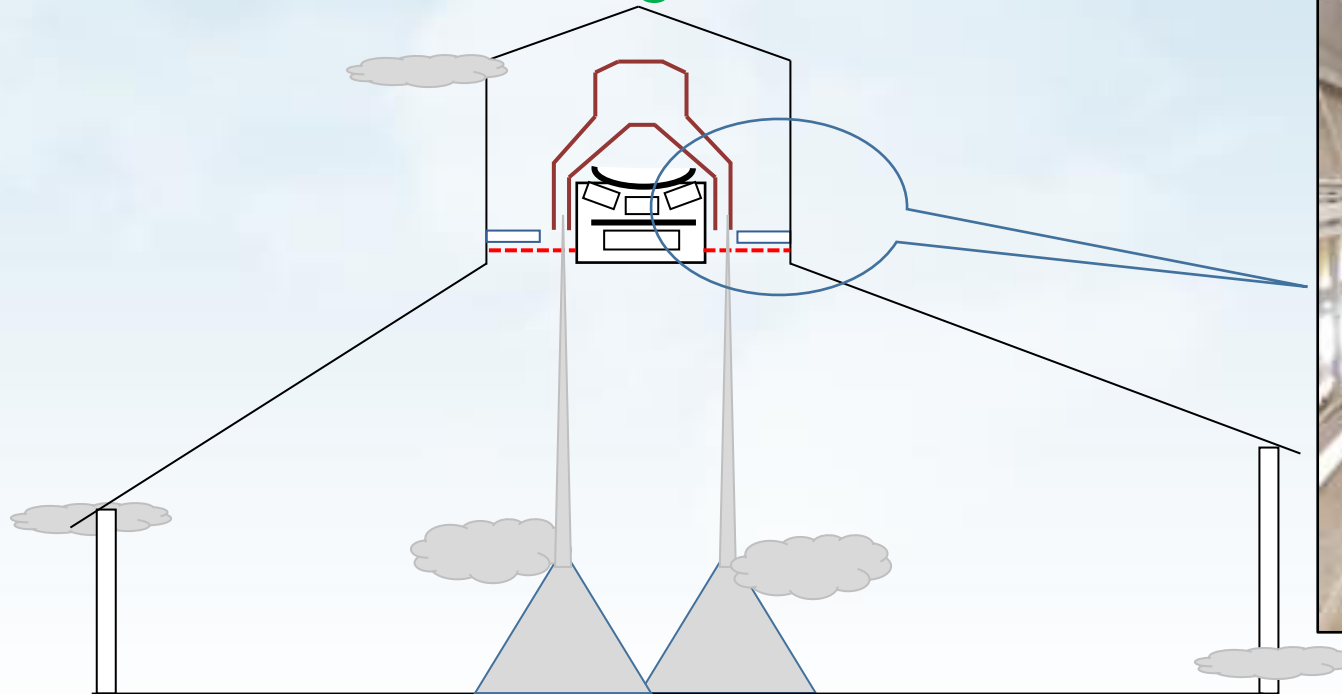
Benefits of Dedusting system for conveyors-

- Clean atmosphere at Transfer Tower points
- Safe working condition near Transfer Points
- Saving of material loss into atmosphere
- Contamination of nearby water bodies is reduced
- Earlier slippery roads were hazardous.

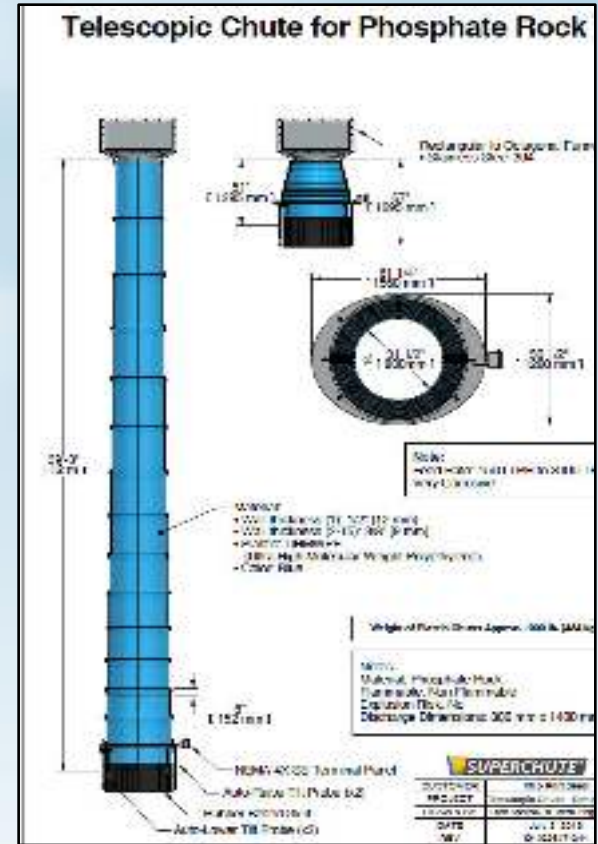
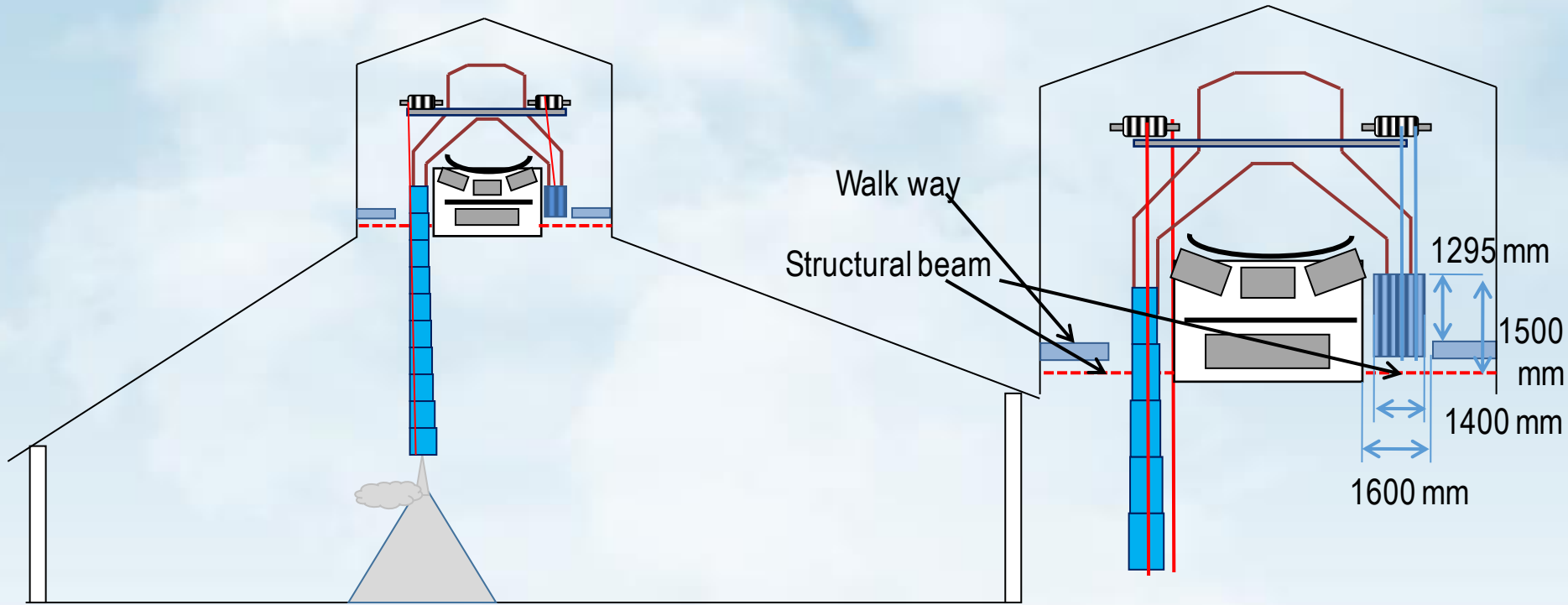
Installation of Retractable type telescopic chute at the tripper

Rock discharge & dust generation

Depiction of rock discharge in silo
and dust generation



Operation of Chute Inside Silo

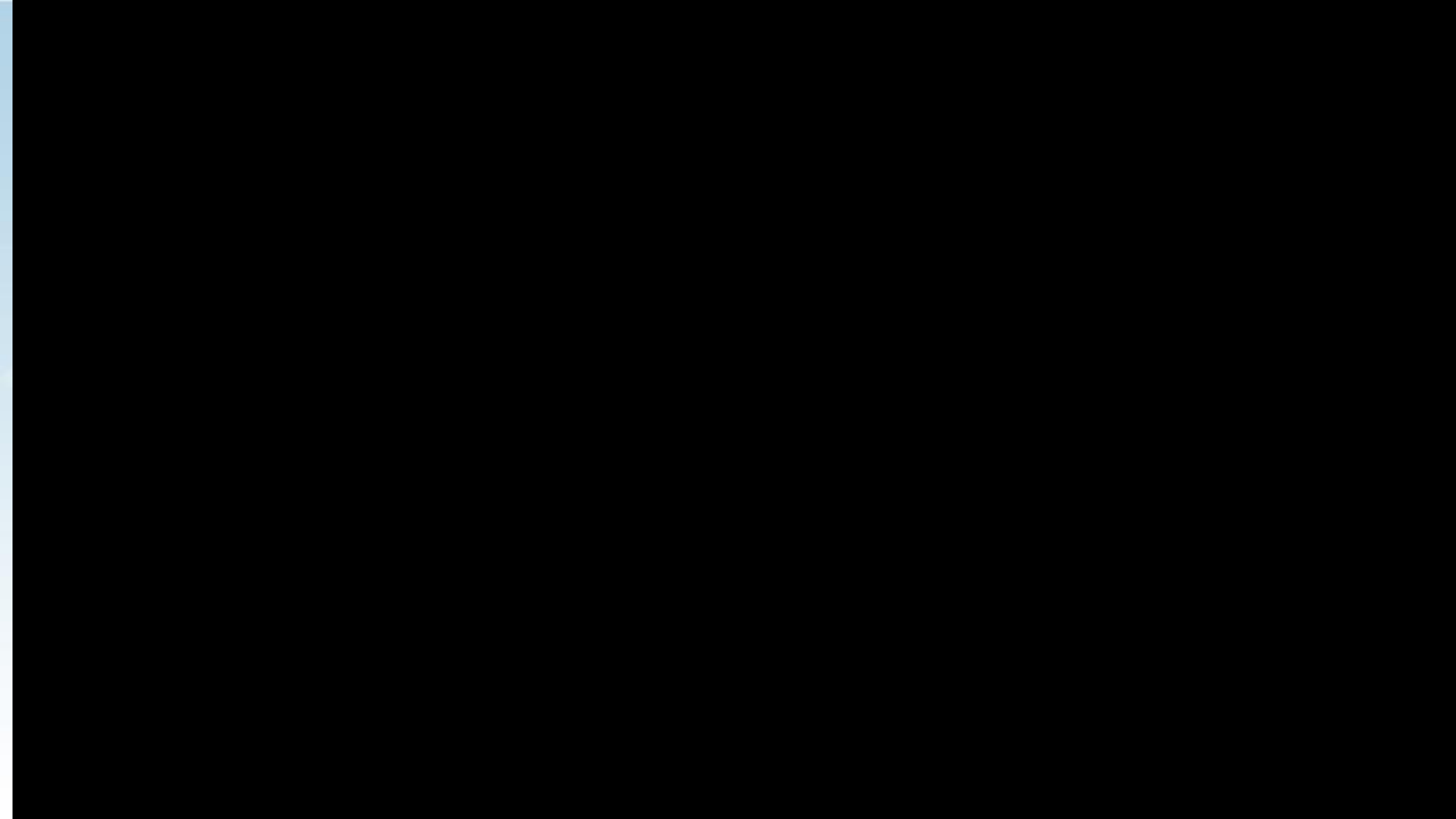


- ❑ Depiction of extended chute dropping material from just above the heap.
- ❑ Chute, when retracted completely, can travel over the structural members.
- ❑ MOC of pipe is Ultra High Molecular Weight PolyEthylene (UHMWPE)

Installation of Retractable type telescopic chute at the tripper

Benefits of Telescopic chute-

- ❑ High dust emission during rock unloading from entrance, side wall, Transfer towers and Maintenance tower got eliminated
- ❑ Environment pollution due to dust emission
- ❑ Loss of material
- ❑ Contamination with Raw Sulphur in adjacent silo & mixing with atmospheric air causing conversion problem in Sulphur Converter
- ❑ Work environment problem - not possible to work inside silo during Rock unloading





Fire Detection alarm system in Sulphur Silo

Control Philosophy / Control Logic of Fire Detection & Alarm System at Sulphur Silo, IFFCO, Paradeep

1. SO₂ gas sensor from any zone detects SO₂ gas and send signal to Fire Alarm Panel through connected monitor module.
2. Fire Alarm Panel send signal to hooters to raise an alarm and activate the respective zone control module for send the signal to DVCP (Deluge Valve Control Panel)
3. LHS Cable detects the heat at 70 DegC and send signal to Fire Alarm Control Panel through connected monitor module.
4. Fire Alarm Panel send signal to hooters to raise an alarm.

The following chart for DV operation with respective SO₂ Sensor activation:

Zone Number	SO ₂ gas detector	DV Number
1	D1, D2,D13,D14,D32	1
2	D2,D3,D14,D15,D31,D32	2
3	D3,D4,D15,D16,D31	3
4	D4,D5,D16,D30	4
5	D5,D6,D17,D18,D29,D30	5
6	D6,D7,D18,D19,D29	6
7	D7,D8,D19,D28	7
8	D9,D20,D21,D27	8
9	D9,D10,D11,D21,D22,D27	9
10	D11,D12,D23,D24,D25,D26	10

Fire Detection alarm system in Sulphur Silo

Sensor installed inside
Sulphur silo



Deluge Valve for Fire
Detection Alarm System

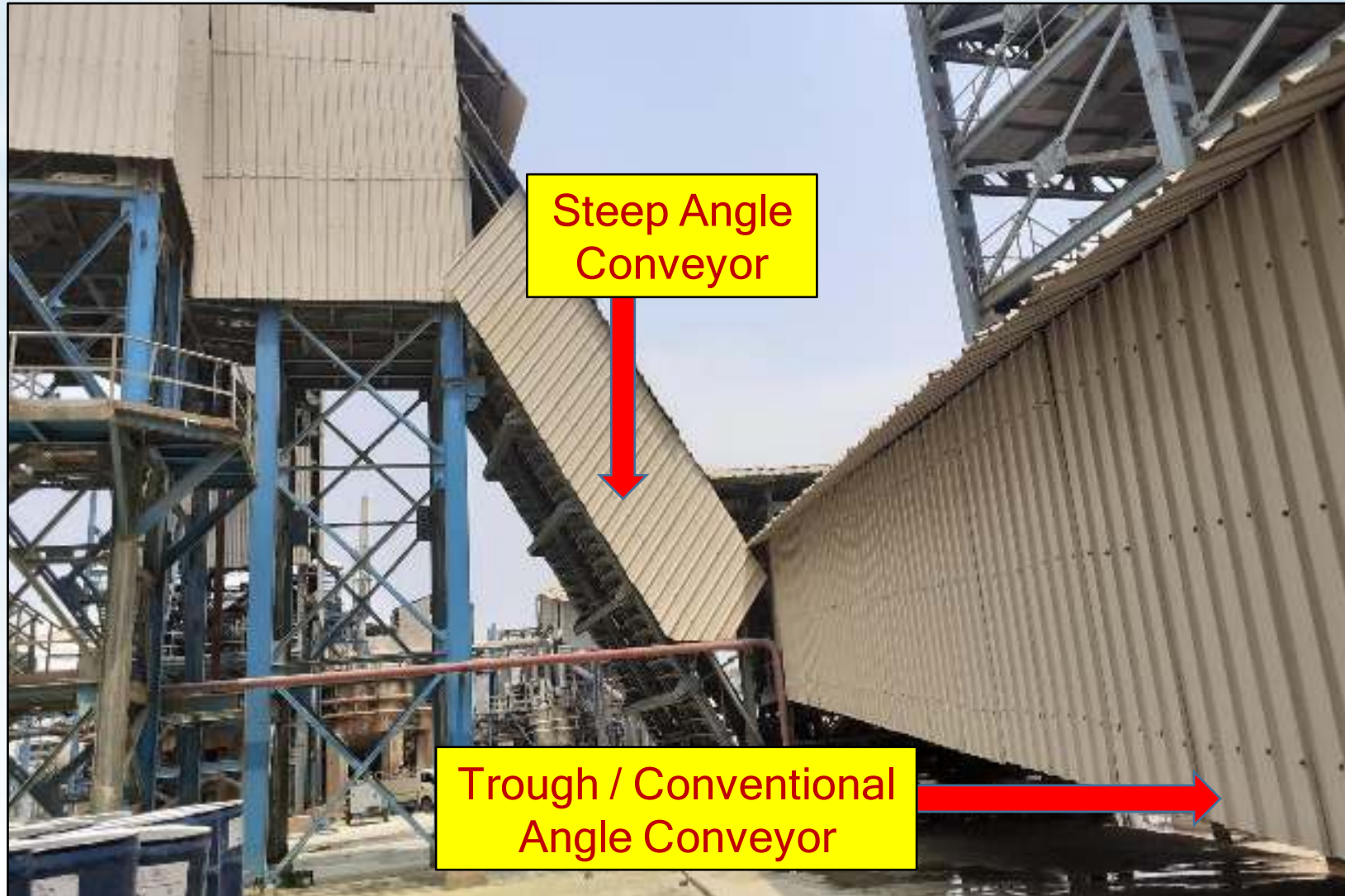


Linear Heat Sensing Cable
installed on Sulphur Conveyor

Sprinkling system inside silo after activation



Installation of Steep Angle Conveyor for sulphur shifting





Steep Angle Belt Conveyor



Trough Angle Belt Conveyor

Salient features & Benefits of the Reclaiming Conveyor

- ❑ Less space is required for its installation
- ❑ Minimum maintenance is required for running the reclaiming conveyor.
- ❑ Present one is 60 deg inclined however 90 deg inclination is also possible.
- ❑ There is no spillage even at such steep inclination

Improvement in Ammonia Tank

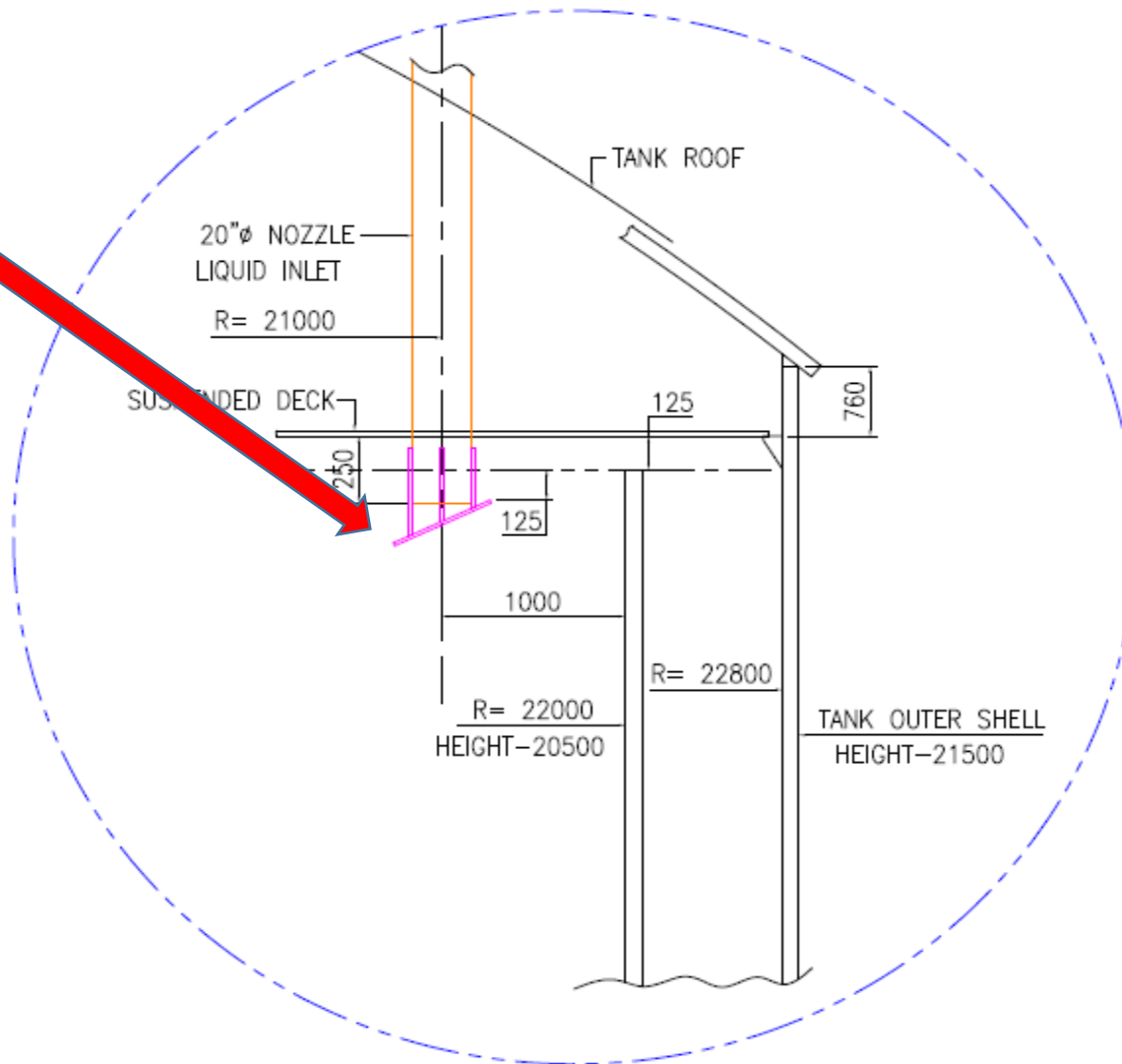
On 1st Nov 2000 a leakage was observed from annular ring plate of Ammonia Tank-B.

The leakage was due to following reasons :-

- ❖ Tremendous external pressure because of liquid ammonia build up in the annular space much above inner tank liquid level for which it was not designed. The deformation of inner tank was due to lifting of inner tank caused by buoyancy.
- ❖ The deformation of inner tank led to failure of anchor plate and sealing plate attached to bottom plate. The liquid ammonia from annular space leaked through sealing plate.
- ❖ The ammonia inlet pipe was only 125 mm below the top of inner tank. The build up in the annular space was only due to splashing of liquid ammonia from discharge end of ammonia inlet pipe during ship unloading operation. (Shown in GA Drawing attached Exploded view-K)

Modification in Ammonia Tank

Inlet line before
modification



DETAIL - "K"
DETAIL BEFORE MODIFICATION

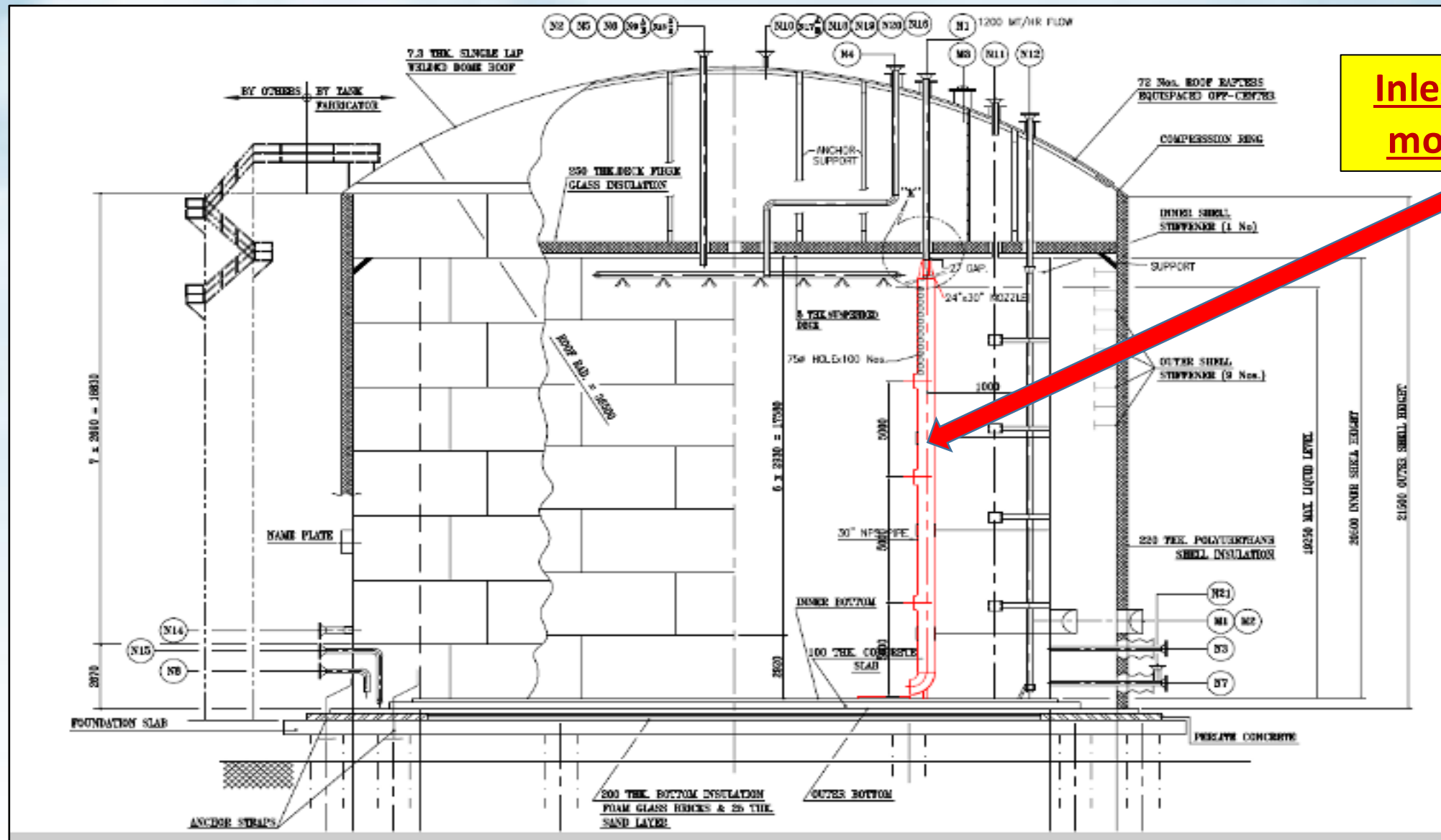
Modification in Ammonia Tank

Modification carried out in Tank-B to avoid recurrence

The following modification was carried out by M/s UHDE India Ltd to avoid re-occurrence:

- ❖ The inlet pipe for ammonia liquid was extended close to bottom of tank (shown in GA drawing).

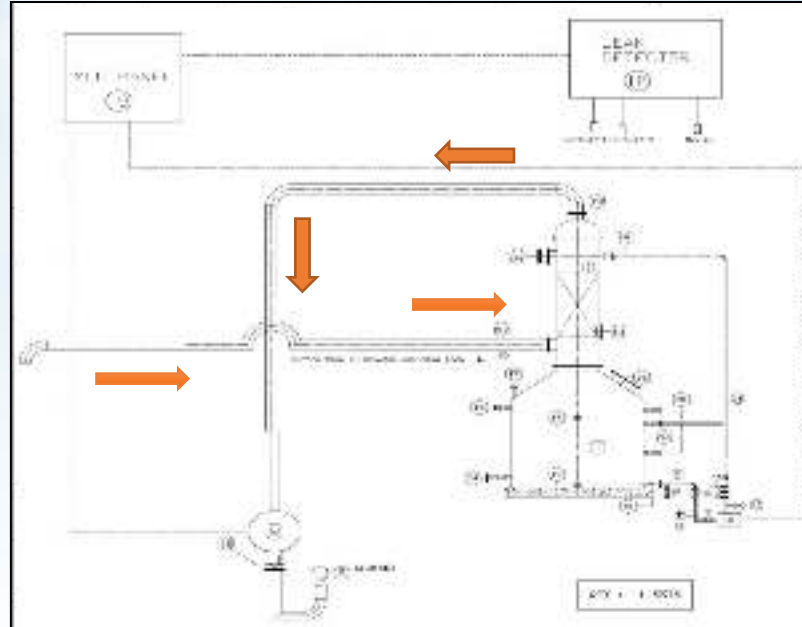
5. Modification in Ammonia Tank



Inlet line after modification

Chlorine Scrubbing System

- ❑ Chlorine is highly hazardous substance.
- ❑ Chlorine Scrubbing System is installed at various location near chlorine tonners.
- ❑ Chlorine from chlorinator is sucked through blower, which is scrubbed in caustic solution.



Condition Monitoring of Stack by using Drone Camera





Overall Improvement in Maintenance Practices / System

Improvement action-1:

Use of Vibration Pen in different maintenance deptt.

Benefits:

- ☐ Checking the instant vibration level of any rotary equipment to ensure the health condition.
- ☐ In case of alarming situation in equipment, immediate preventive maintenance can be taken to avoid accident.



Vibration Pen

Overall Improvement in Maintenance Practices / System

Improvement action-2:

Synthetic Poly urea grease (Omega 85) lubricant applied in all acid circulation pumps:

Previously used lubricant- Servo Gem EP2 gave less bearing life in acidic environment.

Benefits:

- Manpower saving for bearing replacement
- Accidents are reduced
- Bearing life increased
- Pump failure reduced
- Reliability improved.



Omega 85 grease

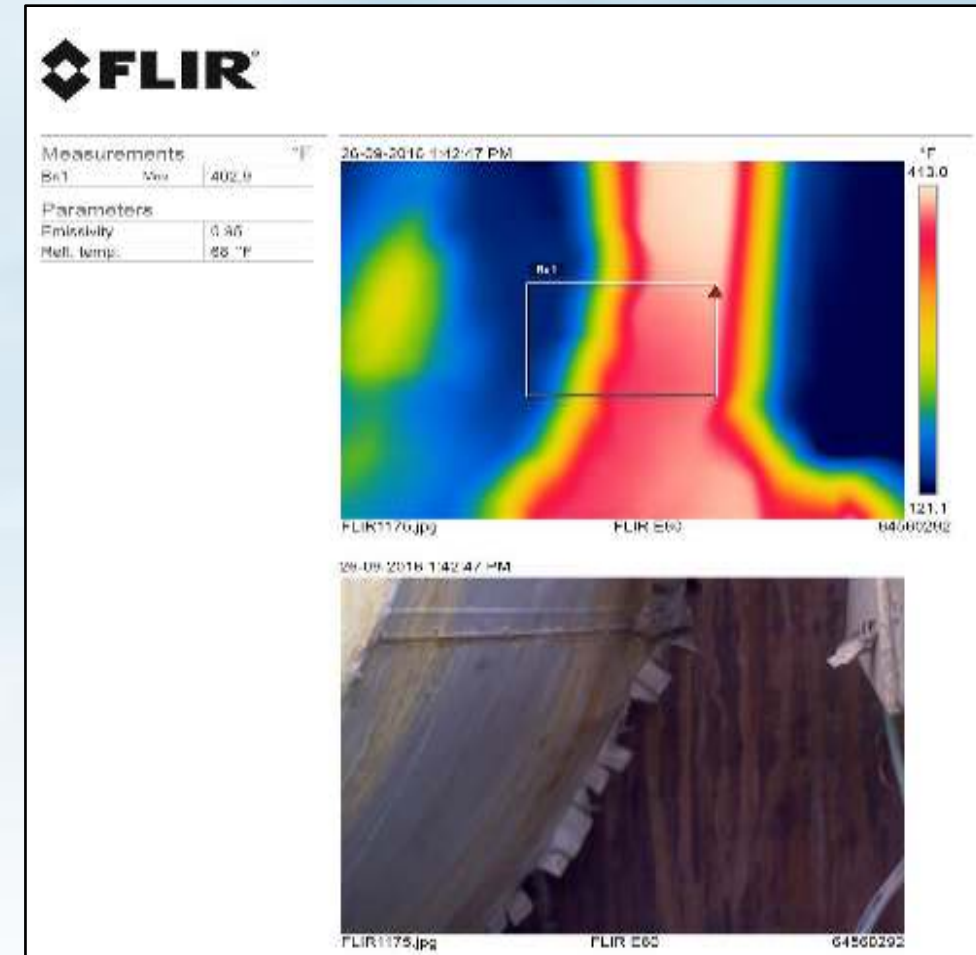
Overall Improvement in Maintenance Practices / System

Improvement action-3:

Thermography survey of shell & dished end of sulphur furnaces in SAP at regular intervals.

Benefits:

To confirm the healthiness of inside brick lining of furnace and in case it was observed that there is drastic increase in surface temperature of furnace, inspection of brick lining can be done to prevent further damage



Thermographic Survey

Overall Improvement in Maintenance Practices / System

Improvement action-5:

- 1) Distributed header brought down which were earlier resting over timber.

Benefits:

- 1) Extra load of pipe on structure reduced, ensuring safety of structure.
- 2) Accidents have been prevented.



Distribution header relocation

Overall Improvement in Maintenance Practices / System

Improvement action-6:

Alignment of Equipment's through Laser Alignment machine.

Benefits:

- Rapid decrease in vibration problem.
- Alignment jobs is being carried out in less time and more effectively & accurately.
- Now almost all technician are confident in carrying out alignment job through laser alignment machine.



EasyLaser Make Laser shaft alignment machine

Overall Improvement in Maintenance Practices / System

Improvement action-7:

Standardization of Hand Railing, Toe/Knee guard

Benefits:

- ❖ Earlier we were using 1½" sch 40 CS pipe / CS angles for hand railing and toe guard purpose. Accidents have been reduced.
- ❖ Instead of using non standardised material for hand railing, we have started using 1½" sch10 SS 304 welded pipes and SS 304 flats of 3 mm thick.
- ❖ Corrosion of material was drastically reduced due to use of SS 304 material.

Common Upgradation in plant

- ❑ MOC Upgradation of Storage Tanks, Pipeline from CSRL to SS 904 / SS 316L. This has reduced leakages & breakdowns.
- ❑ Upgradation from Asbestos AC sheet to Palruf corrugated PVC sheet. This has reduced height job work.
- ❑ MOC Upgradation of Foundation bolts from MS to SS 304L. The health of structure is ensured.
- ❑ Lining in Hoppers with SS 304. Choking of hoppers has greatly reduced.

Robotic Ultrasonic crawler

- ❑ Robotic Ultrasonic crawler SCORPIAN-2 for thickness measurement of tanks, vessels or pipeline. No couplant required.
- ❑ Manual inspection with scaffolding is not required.
- ❑ Time for testing and report generation is very less.
- ❑ Can measure 2.5 mm thick plate to 50 mm.





Best Safety Practices in IFFCO Paradeep Unit

Mobile Application For Incident Reporting

We have developed a mobile application to report Near miss ,Unsafe act & condition. In this mobile app , every employee has authority to report any incident with photo. After reporting, automatic mail is generated to the concerned personnel.



The screenshot displays the 'Unsafe Act' application interface. At the top, there's a title bar with a back arrow and the text 'Unsafe Act'. Below it, a banner for 'IFFCO Farmers Fertiliser Cooperative Ltd. Unsafe Act Photo Uploading System' is visible. A 'Query' section with input fields for 'Date From', 'Date To', and 'Page' is present. The main part of the screen is a table with columns: S.No, Date, Plant, Location, Incident, LAG, Action, Remarks, File name, and Photo. The table contains five rows of incident data.

S.No	Date	Plant	Location	Incident	LAG	Action	Remarks	File name	Photo
25	20/03/2019	SHRI	SLATARA	WALKING	01/0000	Reported	Reported	20190320_100000_000000.jpg	Photo 1
26	20/03/2019	SHRI	SLATARA	WALKING	01/0000	Reported	Reported	20190320_100000_000000.jpg	Photo 1
27	20/03/2019	SHRI	SLATARA	WALKING	01/0000	Reported	Reported	20190320_100000_000000.jpg	Photo 1
28	20/03/2019	SHRI	SLATARA	WALKING	01/0000	Reported	Reported	20190320_100000_000000.jpg	Photo 1
29	20/03/2019	SHRI	SLATARA	WALKING	01/0000	Reported	Reported	20190320_100000_000000.jpg	Photo 1

History Data

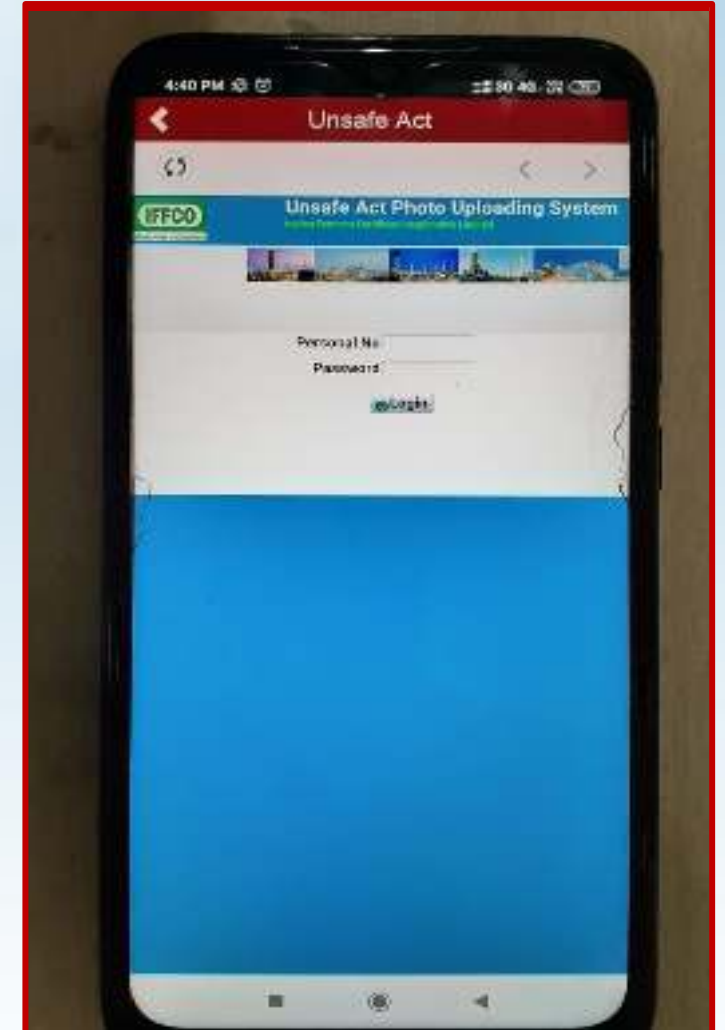


Photo of Mobile app

Use Of Special Tools For Critical Height Jobs

For taking care of safety of workers, we use special tools for height related jobs in the plant. We have fully equipped hydraulically operated Articulated boom, Scissor lift, Palfinger, Truck mounted Hiwa. Specifications of these equipment with their pictures are shown below.

1. Hydraulic Elevated Scissor Lift :

It can move vertically up & down with a maximum height of 16.2 meter and with a maximum weight of 680 Kg Including worker, tools, equipment etc.



Hydraulic Elevated Scissor Lift

Use Of Special Tools For Critical Height Jobs

2. Hydraulically Operated Articulated Boom

It can move vertically as well as horizontally with a Maximum height of 39.5 meter & Maximum weight of 230 Kg including 2 Workmen & Equipment Load



3. Truck Mounted Hiwa

It can move vertically up and down with a maximum height of 13.7 meter and maximum weight = 170 Kg



Use Of Special Tools For Critical Height Jobs

4. Palfinger (Truck Mounted Self Loading Hydraulic Crane)

It has lifting Capacity of 3.84 Ton , Crane Boom Length 8 Meter, Height 12 meter



Safety Hot Spots

We have identified the vulnerable locations in the plant. CCTV cameras have been installed to observe the unsafe act & condition at all critical locations of Ammonia Storage area, Cross Country Conveyor area, Di-Ammonium Phosphate Plant (DAP), Sulphuric Acid Plant (SAP), Phosphoric Acid Plant (PAP). CCTV camera display units are installed in Security control room, Plant Control room and Unit Head chamber. Total **200 no's** of CCTV cameras are installed inside/outside the plant premises.



AAINAA Developed

We have developed following areas in our major plants as “AAINA (Advanced Action in Industries to Abate Accidents)”.

1. Phosphoric Acid Plant **Evaporator ‘G’ & ‘H’ Area**
2. Di-Ammonium Phosphate Plant **Granulator Floor Train -B.**
3. Sulphuric Acid Plant **Acid Area , Train -2**
4. AFBC Boiler , **Boiler Feed Water Pump Area**

In these areas the plant, equipment, housekeeping, machine guards, electrical wiring, means of approach & exit, colour coding, illumination, safety signage etc. are well maintained and complying with statutory provisions and applicable codes.

AINAA Developed



PHOSPHORIC ACID PLANT Area- Evaporator 'G' & 'H'



**DI-AMMONIUM PHOSPHATE PLANT
Area- Granulator Floor**

AINAA Developed



SULPHURIC ACID PLANT Area-Acid Area, Train-2



AFBC BOILER Area- Boiler Feed Water Pump Area



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